Maternal Health Risk Level Prediction Using Machine Learning

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GitHub repository link :

1. Problem Definition

Quantitatively assessing maternal physiological status during pregnancy and identifying high-risk groups at an early stage is one of the key challenges in improving obstetric outcomes. Vital signs such as blood pressure, blood glucose, heart rate, and body temperature serve as early indicators of pregnancy-related complications. Developing automated risk prediction models based on these indicators can support more efficient allocation of medical resources and the formulation of timely intervention strategies. This project aims to build a multi-class classification model that categorizes maternal health status into Low Risk, Mid Risk, and High Risk using a publicly available clinical dataset. Additionally, model interpretability techniques are employed to ensure clinical reliability and transparency of the predictions.

2. Dataset Description

The dataset used in this study is the publicly available Maternal Health Risk Dataset from Kaggle. It includes physiological indicators of pregnant women, with the target label RiskLevel classified into three categories—Low Risk, Mid Risk, and High Risk—based on medical expert assessments. The dataset contains six independent variables: Age, SystolicBP, DiastolicBP, BS (blood sugar), BodyTemp, and HeartRate.

3. Data Preprocessing

3.1 Duplicate Removal & Missing Value Handling

Approximately half of the dataset contained duplicate rows, which were removed to prevent bias during model training caused by repeated records from the same patient.The dataset was examined for missing values, but none were found; thus, no additional imputation or interpolation was required.

3.2 Outlier Filtering Based on Clinical Bounds

In medical datasets, extreme values may hold clinical significance and indicate potential conditions, making statistical methods. Therefore, we opted to filter out only clinically implausible values based on physiological bounds defined by the WHO and ACOG (e.g., reproductive age range of 15–49 years, normal blood pressure, and body temperature). This approach helped eliminate values that are realistically impossible rather than statistically rare.

3.3 Skewness Correction

Variables that strongly violate normality assumptions can lead to biased weight estimation in machine learning models. To mitigate this, we corrected skewness using the following methods:

- **BS** : Applied Box-Cox transformation to reduce positive skew.
- **Age**: Used Yeo-Johnson transformation to address negative skew.

For **BodyTemp**, since the values were highly concentrated in a narrow range, the transformation was deemed ineffective and therefore omitted.

3.4 Label Encoding & Feature Scaling

The target variable, RiskLevel, was composed of string labels and was converted into integer classes (0, 1, 2) using LabelEncoder to make it suitable for model input. All numerical input features were scaled to a standard normal distribution (mean 0, variance 1) using StandardScaler.

4. Model Description

In this project, we aimed to classify maternal health risk levels (Low, Mid, High) based on physiological indicators by comparing the performance of various machine learning classifiers. Specifically, three representative models were selected: (1) K-Nearest Neighbors (KNN), a simple distance-based algorithm useful as a baseline; (2) Random Forest (RF), a robust ensemble method known for stability and interpretability through feature importance; and (3) XGBoost (XGB), a gradient-boosting model with strengths in regularization and handling complex nonlinear relationships.

Model selection was grounded in their complementary characteristics: KNN for its simplicity and sensitivity to data distribution, RF for its high accuracy and feature-level transparency, and XGB for its strong performance in structured datasets. During training, we observed a class imbalance ratio of approximately 2:1:1 for Low:Mid:High risk levels. To mitigate bias and enhance generalization for minority classes (Mid and High), SMOTE (Synthetic Minority Over-sampling Technique) was incorporated into the preprocessing pipeline.

All models were evaluated under consistent preprocessing and validation conditions. Hyperparameter tuning was conducted using GridSearchCV, and fairness in comparison was ensured through Stratified 5-Fold Cross Validation.

4. Evaluation Metrics & Methods

The performance of the models was evaluated using three key metrics:

- **Accuracy**: A basic metric representing the proportion of correctly classified instances across the entire dataset.
- **F1-Macro Score**: The unweighted average of F1-scores across all classes, providing a balanced evaluation of performance in the presence of class imbalance—particularly useful when mid and high risk classes are underrepresented.
- **Multi-Class ROC-AUC**: Computed by calculating the ROC curve for each class against all others and then averaging the results, this metric evaluates the model's ability to distinguish between classes.

This multi-metric evaluation is crucial for assessing whether a model performs uniformly across all risk categories without overfitting to the dominant class. In particular, the F1-macro score captures the balance between precision and recall for each risk group (Low, Mid, High), while the ROC-AUC metric provides insight into the model's confidence and its ability to clearly separate class boundaries.

5. Results & Interpretation

Model performance was assessed based on three metrics—Accuracy, F1-Macro, and ROC-AUC—derived through Stratified 5-Fold Cross-Validation.

	KNN	RF	XGB
Accuracy	0.577	0.699	0.673
ROC-AUC	0.724	0.781	0.773
F1-Macro	0.549	0.646	0.615

In addition to performance metrics, model interpretability was addressed using SHAP-based explanation. SHAP Summary Plots for each class revealed that SystolicBP, DiastolicBP, and BS were the most influential features contributing to prediction outcomes.

Additionally, local interpretation using SHAP Waterfall Plots was performed for individual patients, visually explaining the key features that contributed to specific prediction outcomes.

6. Development Environment

- **OS**: Google Colab (Linux-based)
- **Python version**: 3.11
- **Key Libraries**: scikit-learn, xgboost, imblearn, shap, matplotlib, seaborn, pandas, numpy